Proposed Prescribing Information For the use of a Registered Medicinal Practitioner or a Hospital or a Laboratory only

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DAPTON GEL (DAPSONE GEL, 7.5%)

COMPOSITION

Dapton Gel
Each gram of gel contains:
Dapsone USP ………………………..75 mg
Gel Base………………………………..q.s.

Preservative:
Methyl Paraben……………………….2 mg .

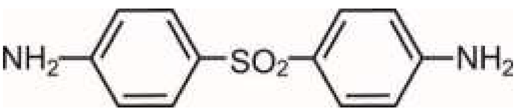
Dosage from
Gel

INDICATION

Dapton (dapsone) Gel, 7.5%, is indicated for the topical treatment of acne vulgaris in patients 12 years of age and older.

DESCRIPTION

Dapton (dapsone) Gel, 7.5%, contains dapsone, a sulfone, in an aqueous gel base for topical dermatologic use. Dapsone Gel, 7.5%, is an off-white to yellow colour homogeneous gel with suspended particles, no phase separation and sedimentation, free of lumps and foreign matter. Chemically, dapsone has an empirical formula of $C_{12}H_{12}N_2O_2S$. It is a white to creamy white crystalline powder that has a molecular weight of 248.30. Dapsone's chemical name is 4-[(4-aminobenzene) sulfonyl] aniline and its structural formula is:



Each gram of Dapsone Gel, 7.5%, contains 75 mg of dapsone, USP, in a gel of diethylene glycol monoethyl ether, methylparaben, carbomer homopolymer type C, sodium hydroxide and purified water.

DOSE AND METHOD OF ADMINISTRATION

For topical use only. Not for oral, ophthalmic, or intravaginal use.

After the skin is gently washed and patted dry, apply approximately a pea-sized amount of Dapton Gel, 7.5%, in a thin layer to the entire face once daily. In addition, a thin layer may be applied to other affected areas once daily. Rub in Dapton Gel, 7.5%, gently and completely.

If there is no improvement after 12 weeks, treatment with Dapton Gel, 7.5%, should be reassessed.

CONTRAINDICATIONS

None.

SPECIAL WARNINGS AND PRECAUTIONS

Hematological Effects

Methemoglobinemia

Cases of methemoglobinemia, with resultant hospitalization, have been reported post-marketing in association with twice daily dapsone gel, 5%, treatment. Patients with glucose-6-phosphate dehydrogenase deficiency or congenital or idiopathic methemoglobinemia are more susceptible to drug-induced methemoglobinemia. Avoid use of Dapsone Gel, 7.5%, in those patients with congenital or idiopathic methemoglobinemia.

Signs and symptoms of methemoglobinemia may be delayed some hours after exposure. Initial signs and symptoms of methemoglobinemia are characterized by a slate grey cyanosis seen in, e.g., buccal mucous membranes, lips, and nail beds. Advise patient to discontinue Dapsone Gel, 7.5%, and seek immediate medical attention in the event of cyanosis.

Dapsone can cause elevated methemoglobin levels particularly in conjunction with methemoglobin-including agents [*see Drug Interactions*].

Hemolysis

Oral dapsone treatment has produced dose-related hemolysis and hemolytic anemia. Individuals with glucose-6-phosphate dehydrogenase (G6PD) deficiency are more prone to hemolysis with the use of certain drugs. G6PD deficiency is most prevalent in populations of African, South Asian, Middle Eastern, and Mediterranean ancestry.

In clinical trials, there was no evidence of clinically relevant hemolysis of hemolytic anemia in subjects treated with topical dapsone. Some subjects with G6PD deficiency using dapsone gel, 5%, twice daily developed laboratory changes suggestive of hemolysis [*see Use in Specific Populations*].

Discontinue Dapsone gel, 7.5%, if signs and symptoms suggestive of hemolytic anemia occur. Avoid use of Dapsone gel, 7.5%, in patients who are taking oral dapsone or antimalarial medications because of the potential for hemolytic reactions. Combination of Dapsone gel, 7.5%, with trimethoprim/sulfamethoxazole (TMP/SMX) may increase the likelihood of hemolysis in patients with G6PD deficiency [*see Drug Interactions*].

Peripheral Neuropathy

Peripheral neuropathy (motor loss and muscle weakness) has been reported with oral dapsone treatment. No events of peripheral neuropathy were observed in clinical trials with topical dapsone treatment.

Skin Reactions

Skin reactions (toxic epidermal necrolysis, erythema multiforme, morbilliform and scarlatiniform reactions, bullous and exfoliative dermatitis, erythema nodosum, and urticaria) have been reported with oral dapsone treatment. These types of skin reactions were not observed in clinical trials with topical dapsone treatment.

ADVERSE REACTIONS

Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice.

A total of 2,161 subjects were treated with Dapsone gel, 7.5%, for 12 weeks in 2 controlled clinical trials. The population ranged in age from 12 to 63 years, was 56% female, and 58% Caucasian. Adverse drug reactions that were reported in at least 0.9% of subjects treated with Dapsone gel, 7.5%, appear in Table 1 below.

Table 1 Adverse Reactions Occurring in at Least 0.9% of Subjects with Acne Vulgaris in 12-week Controlled Clinical Trials

	Dapsone Gel, 7.5% (N=2,161)	**Vehicle (N=2,175)**
Application Site Dryness	24 (1.1%)	21 (1.0%)
Application Site Pruritus	20 (0.9%)	11 (0.5%)

Experience with Oral Use of Dapsone

Although not observed in the clinical trials with topical dapsone, serious adverse reactions have been reported with oral use of dapsone, including agranulocytosis, hemolytic anemia, peripheral neuropathy (motor loss and muscle weakness), and skin reactions (toxic epidermal necrolysis, erythema multiforme, morbilliform and scarlatiniform reactions, bullous and exfoliative dermatitis, erythema nodosum, and urticaria).

Post-marketing Experience

The following adverse reactions have been identified during post-approval use of topical dapsone. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

Methemoglobinemia has been identified during postmarketing use of topical dapsone [*see Warnings and Precautions*].

DRUG-INTERACTION

No formal drug-drug interaction studies were conducted with Dapsone gel, 7.5%.

Trimethoprim– Sulfamethoxazole

A drug-drug interaction study evaluated the effect of the use of dapsone gel, 5%, in combination with double strength (160 mg/800 mg) trimethoprim-sulfamethoxazole (TMP/SMX). During co-administration, systemic levels of TMP and SMX were essentially unchanged, however, levels of dapsone and its metabolites increased in the presence of TMP/SMX. The systemic exposure from Dapsone gel, 7.5%, is expected to be about 1% of that from the 100 mg oral dose, even when co-administered with TMP/SMX.

Topical Benzoyl Peroxide

Topical application of dapsone gel followed by benzoyl peroxide in patients with acne vulgaris may result in a temporary local yellow or orange discoloration of the skin and facial hair.

Drug Interactions with Oral Dapsone

Certain concomitant medications (such as rifampin, anticonvulsants, St. John's wort) may increase the formation of dapsone hydroxylamine, a metabolite of dapsone associated with hemolysis. With oral dapsone treatment, folic acid antagonists such as pyrimethamine have been noted to possibly increase the likelihood of hematologic reactions.

Concomitant Use with Drugs that Induce Methemoglobinemia

Concomitant use of Dapsone gel, 7.5%, with drugs that induce methemoglobinemia such as sulfonamides, acetaminophen, acetanilide, aniline dyes, benzocaine, chloroquine, dapsone, naphthalene, nitrates and nitrites, nitrofurantoin, nitroglycerin, nitroprusside, pamaquine, para-aminosalicylic acid, phenacetin, phenobarbital, phenytoin, primaquine, and quinine may increase the risk for developing methemoglobinemia *[see Warnings and Precautions].*

USE IN SPECIFIC POPULATIONS

Pregnancy

Teratogenic Effects: Pregnancy Category C

There are no adequate and well controlled studies in pregnant women. Dapsone gel, 7.5%, should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus. Dapsone has been shown to have an embryocidal effect in rats and rabbits when administered orally during the period of organogenesis in doses of 75 mg/kg/day and 150 mg/kg/day, respectively (approximately 1400 and 425 times, respectively, the systemic exposure that is associated with the maximum recommended human dose (MRHD) of Dapsone gel, 7.5%, based on AUC comparisons). These effects may have been secondary to maternal toxicity.

Nursing Mothers

Although systemic absorption of dapsone following topical application of Dapsone gel, 7.5%, is minimal relative to oral dapsone administration, it is known that dapsone is excreted in human milk. Because of the potential for oral dapsone to cause adverse reactions in nursing infants, a decision should be made whether to discontinue nursing or to discontinue Dapsone gel, 7.5%, taking into account the importance of the drug to the mother.

Pediatric Use

Safety and efficacy was evaluated in 1066 subjects aged 12-17 years old treated with Dapsone gel, 7.5%, in the clinical trials. The safety profile for Dapsone gel, 7.5%, was similar to the vehicle control group. Safety and effectiveness of Dapsone gel, 7.5%, have not been established in pediatric patients below the age of 12 years.

Geriatric Use

Clinical trials of Dapsone gel, 7.5%, did not include sufficient numbers of subjects aged 65 years and over to determine whether they respond differently from younger subjects.

Glucose-6-phosphate Dehydrogenase (G6PD) Deficiency

Individuals with glucose-6-phosphate dehydrogenase (G6PD) deficiency may be more prone to methemoglobinemia and hemolysis [*see Warnings and Precautions*].

Dapsone gel, 5%, and vehicle were evaluated in a randomized, double-blind, cross-over design clinical study of 64 subjects with G6PD deficiency and acne vulgaris. Subjects were Black (88%), Asian (6%), Hispanic (2%) or of other racial origin (5%). Blood samples were taken at Baseline, Week 2, and Week 12 during both vehicle and Dapsone Gel, 5% treatment periods. Some of these subjects developed laboratory changes suggestive of hemolysis, but there was no evidence of clinically significant hemolytic anemia in this study [*see Warnings and Precautions*].

CLINICAL PHARMACOLOGY

Mechanism of Action

The mechanism of action of dapsone gel in treating acne vulgaris is not known.

PHARMACOLOGICAL PROPERTIES

Pharmacokinetic properties

In a pharmacokinetic study, male and female subjects 16 years of age and older with acne vulgaris (N=19) applied 2 grams of Dapsone Gel, 7.5%, to the face, upper chest, upper back and shoulders once daily for 28 days. Steady state for dapsone was reached within 7 days of dosing. On day 28, the mean dapsone maximum plasma concentration (Cmax) and area under the concentration-time curve from 0 to 24 hours post dose ($AUC_{0\text{-}24h}$) were 13.0 ± 6.8 ng/mL and 282 ± 146 ng•h/mL, respectively. The systemic exposure from Dapsone Gel, 7.5%, is expected to be about 1% of that from a 100 mg oral dose.

Long-term safety studies were not conducted with Dapsone Gel, 7.5%, however, in a long-term clinical study of dapsone gel, 5% treatment (twice daily), periodic blood samples were collected up to 12 months to determine systemic exposure of dapsone and its metabolites in approximately 500 subjects. Based on the measurable dapsone concentrations from 408 subjects (M=192, F=216), obtained at Month 3, neither gender nor race appeared to affect the pharmacokinetics of dapsone. Similarly, dapsone exposures were approximately the same between the age groups of 12 to 15 years (N=155) and those greater than or equal to 16 years (N=253). There was no evidence of increasing systemic exposure to dapsone over the study year in these subjects.

Microbiology

In Vivo Activity: No microbiology or immunology studies were conducted during Dapsone Gel, 7.5%, clinical trials.

Drug Resistance: No dapsone resistance studies were conducted during dapsone gel clinical studies. Because no such studies were done, there are no data available as to whether dapsone treatment may have resulted in decreased susceptibility of *Propionibacterium acnes*, an organism associated with acne, or to other antimicrobials that may be used to treat acne. Therapeutic resistance to dapsone has been reported for *Mycobacterium leprae*, when patients have been treated with oral dapsone.

NONCLINICAL TOXICOLOGY

Carcinogenesis, Mutagenesis, Impairment of Fertility

Dapsone was not carcinogenic to rats when orally administered to females for 92 weeks or males for 100 weeks at dose levels up to 15 mg/kg/day (approximately 340 times the systemic exposure observed in humans as a result of use of the MRHD of Dapsone Gel, 7.5%, based on AUC comparisons).

No evidence of potential to induce carcinogenicity was observed in a dermal study in which dapsone gel was topically applied to Tg.AC transgenic mice for approximately 26 weeks. Dapsone concentrations of 3%, 5%, and 10% were evaluated; 3% material was judged to be the maximum tolerated dosage.

Topical gels that contained dapsone at concentrations up to 5% did not increase the rate of formation of ultraviolet light-induced skin tumors when topically applied to hairless mice in a 12-month photocarcinogenicity study.

Dapsone was not mutagenic in a bacterial reverse mutation assay (Ames test) using *S. typhimurium* and *E. coli*, with and without metabolic activation, and was negative in a micronucleus assay conducted in mice. Dapsone increased both numerical and structural aberrations in a chromosome aberration assay conducted with Chinese hamster ovary (CHO) cells.

The effects of dapsone on fertility and general reproduction performance were assessed in male and female rats following oral (gavage) dosing. Dapsone reduced sperm motility at dosages of 3 mg/kg/day or greater (approximately 22 times the systemic exposure that is associated with the MRHD of Dapsone Gel, 7.5%, based on AUC comparisons). The mean numbers of embryo implantations and viable embryos were significantly reduced in untreated females mated with males that had been dosed at 12 mg/kg/day or greater (approximately 187 times the systemic exposure that is associated with the MRHD of ACZONE Gel, 7.5%, based on AUC comparisons), presumably due to reduced numbers or effectiveness of sperm, indicating impairment of fertility. Dapsone had no effect on male fertility at dosages of 2 mg/kg/day or less (approximately 15 times the systemic exposure that is associated with the MRHD of Dapsone Gel, 7.5%, based on AUC comparisons). When administered to female rats at a dosage of 75 mg/kg/day (approximately 1400 times the systemic exposure that is associated with the MRHD of Dapsone Gel, 7.5%, based on AUC comparisons) for 15 days prior to mating and for 17 days thereafter, dapsone reduced the mean number of implantations, increased the mean early resorption rate, and reduced the mean litter size. These effects were probably secondary to maternal toxicity.

Dapsone was assessed for effects on perinatal/postnatal pup development and postnatal maternal behavior and function in a study in which dapsone was orally administered to female rats daily beginning on the seventh day of gestation and continuing until the twenty-seventh day postpartum. Maternal toxicity (decreased body weight and food consumption) and developmental effects (increase in stillborn pups and decreased pup weight) were seen at a dapsone dose of 30 mg/kg/day (approximately 560 times the systemic exposure that is associated with the MRHD of Dapsone Gel, 7.5%, based on AUC comparisons). No effects were observed on the viability, physical development, behavior, learning ability, or reproductive function of surviving pups.

CLINICAL STUDIES

The safety and efficacy of once daily use of Dapsone Gel, 7.5%, was assessed in two 12-weeks multicenter, randomized, double-blind, vehicle-controlled trials. Efficacy was associated in a total of 4340 subjects 12 years of age and older. The majority of the subjects had moderate acne vulgaris, 20 to 50 inflammatory and 30 to 100 non-inflammatory lesions at baseline, who were randomized to receive either Dapsone Gel, 7.5%, or vehicle.

Treatment response was defined at Week 12 as the proportion of subjects who were rated “none” or “minimal” with at least a two-grade improvement from baseline on the Global Acne Assessment Score (GAAS), and mean absolute change from baseline in both inflammatory and non-inflammatory lesion counts. A GAAS score of “none” corresponded to no evidence of facial acne vulgaris. A GAAS score of “minimal” corresponded to a few non-inflammatory lesions (comedones) being present and to a few inflammatory lesions (papules/pustules) that may be present.

The GAAS success rate, mean reduction, and percent reduction in acne lesion counts from baseline after 12 weeks of treatment are presented in the following table.

Table 2 Clinical Efficacy of Dapsone Gel at Week 12 in Subjects with Acne Vulgaris

	Trial 1		Trial 2	
	Dapsone Gel, 7.5% (N=1044)	**Vehicle (N=1058)**	**Dapsone Gel, 7.5% (N=1118)**	**Vehicle (N=1120)**
Global Acne Assessment Score				
GAAS Success (Score 0 or 1)	30%	21%	30%	21%
Inflammatory Lesions				
Mean absolute reduction	16.1	14.3	15.6	14.0
Mean percent reduction	56%	49%	54%	48%
Non-inflammatory Lesions				
Mean absolute reduction	20.7	18.0	20.8	18.7
Mean percent reduction	45%	39%	46%	41%

HOW SUPPLIED/PRESENTATION

Dapsone Gel is an off-white to yellow colour homogeneous gel with suspended particles; no phase separation and sedimentation; free of lumps and foreign matter. It is supplied in an airless polypropylene pump containing a polypropylene bottle with a high-density polyethylene piston.

Dapton Gel, 7.5%, is supplied in the 60 gram pump and 90 gram pump.

STORAGE CONDITION

Store below 30°C. Protect from freezing.

PATIENT COUNSELING INFORMATION

Advise the patient to read the DCA-approved patient labeling (Patient Information).

Hematological Effects

- Inform patients that methemoglobinemia can occur with topical dapsone treatment. Advise patients to seek immediate medical attention if they develop cyanosis [*see Warnings and Precautions*].

- Inform patients who have G6PD deficiency that hemolytic anemia may occur with topical dapsone treatment. Advise patients to seek medical attention if they develop signs and symptoms suggestive of hemolytic anemia [*see Warnings and Precautions*].

Important Administration Instructions

- Advise patients to apply Dapsone Gel, 7.5%, once daily to the entire face [see Dosage and Administration (2)].
- Dapsone Gel, 7.5% is for topical use only.
- Do not apply Dapsone Gel, 7.5% to eyes, mouth, or mucous membranes.

DATE OF REVISION OF PACKAGE INSERT
03/05/2025

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